Recap Checkpoint 4 — Cumulative: 1.1–1.5 + 2.1 (Computational Thinking)

Name: ______________________ Date: __________ Mark: ______ / 30

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). *(1.1 — Processors)* State the three phases of the **fetch–decode–execute cycle** and, for the fetch phase, name **one** register whose contents are used and **one** register that is updated.

Q2 [4] (AO2). *(1.4 — Data representation)* Show all working.

(a) Convert the denary number **205** to **8-bit binary**. **[1]**

(b) Convert **205** to **hexadecimal**. **[2]**

(c) State **one** reason programmers prefer hexadecimal to binary when reading memory dumps. **[1]**

Q3 [4] (AO2). *(1.4 — Two's complement)* Using **8-bit two's complement**, calculate **38 − 53**.

(a) Show 38 and 53 in 8-bit binary. **[1]**

(b) Form the two's complement of 53. **[1]**

(c) Perform the addition and state the final **denary** result. **[2]**

Q4 [3] (AO1/AO2). *(1.2 — Software development)* A team is building software where requirements are expected to change frequently and the customer wants working software early.

(a) Name a suitable **development methodology**. **[1]**

(b) Give **two** reasons it suits this situation. **[2]**

Q5 [4] (AO1). *(1.3 — Exchanging data / networks)* A file is sent across the Internet using the **TCP/IP stack**.

(a) Name the **four layers** of the TCP/IP stack in order from top to bottom. **[2]**

(b) State the layer at which **packets are routed** between networks and name the layer that **adds port numbers**. **[2]**

Q6 [3] (AO3). *(1.5 — Legal/ethical)* A company plans to use AI to monitor employees' keystrokes to measure productivity.

Discuss **one legal** and **one ethical** issue this raises. **[3]**

Q7 [4] (AO2). *(2.1 — Computational thinking)* A weather app requests live temperature data for thousands of users every few seconds.

(a) Explain how **caching** the data on the server improves performance. **[2]**

(b) State **one** trade-off of caching in this scenario. **[1]**

(c) State what is meant by *abstraction* in the context of modelling the weather. **[1]**

Q8 [5] (AO3). *(2.1 synoptic, extended response)* A car-park company is building software to control entry/exit barriers, track free spaces and serve a mobile app showing live availability.

Explain how **decomposition** and **abstraction** would help the developers manage this problem, and identify **one** part of the system best suited to a **concurrent** (parallel) approach. **[5]**